Casey Barton

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PROFILE

I am a Staff-level operator who works where customer urgency, technical ambiguity, and product execution intersect: diagnose the real failure, ship the production fix, and turn the learning into a reusable pattern. My practice is governed agent engineering — fail-closed deploy gates, root-cause quarantine, eval regression gates that block regressions from shipping, and governed multi-agent systems backed by **65** authored MCP connectors with typed contracts and audit receipts. I work in the open, in writing, with production discipline.

Three years as a founder running an evidence-grade agent control plane from Honolulu: hand-rolled multi-agent loops, retrieval, memory, guardrails, and typed schema validation on every tool contract; full run traces so any execution can be reconstructed; per-task latency and token-cost instrumentation; cross-layer diagnosis across application code, data stores, APIs, infrastructure, and observability. Python primary, strong APIs/automation, production code discipline with review-level standards.

CORE COMPETENCIES

Field engineering: root-cause diagnosis across app / data / API / infra · adoption-blocker resolution · fail-closed deploy gates · eval regression gates · root-cause quarantine · technical stewardship · coordination without formal authority · written technical narratives & reference patterns

AI & Evaluation: agentic workflows · AI governance & auditability · model evaluation · context management · prompt & workflow design · human-in-the-loop delivery · LLM evaluation harnesses · quality scoring · LLM-as-a-judge with written rubrics · golden datasets · behavioral assertions · trace intelligence · parallel tool orchestration · failure-mode instrumentation · MCP servers & connectors · structured outputs · guardrails · fail-closed recovery

Production: Python (primary) · TypeScript · SQL · Bash · REST APIs · JSON Schema / Pydantic-style validation · GitHub Actions (CI/CD) · Postgres · webhooks · streaming workflows · containers · observability & run logging · token budgets · retry logic · latency/cost benchmarking · documentation engineers actually read

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Diagnose and resolve high-impact adoption blockers across complex deployments: systematic root-cause analysis across application code, data stores, APIs, infrastructure, and observability — with root-cause quarantine so fixes address causes, not symptoms.
- Operate fail-closed deploy gates: eval regression gates that block a regression from shipping; CI checks that refuse promotion without evidence; approval gates on every consequential action — governance as the delivery path, not overhead.
- Ship governed AI-adoption patterns: AI governance, auditability, and human-in-the-loop delivery across **65** MCP connector integrations — typed contracts, structured validation on every tool interface, evidence boundaries, and audit receipts on every governed action.

- Turn field learning into durable product: translate failure modes found in production into reusable implementation patterns, reference architectures, runbooks, and documentation — outcomes designed to transfer to long-term owners, not create dependency.
- Design agent systems for failure: instrument stuck loops, bad retrievals, tool misuse, and context blowouts — then build the catchers: replay loops, fail-closed gates, behavioral assertions, and recovery restricted to evidence declared before reassessment.
- Drive reliability, latency, and cost tradeoffs with measured data: per-task latency and token-cost instrumentation on every run; model selection per task complexity; trace intelligence that names the slowest calls and redundant tool use.
- Run an evidence-grade agent control plane across a **1,395-repository** estate: typed handoff contracts, guardrails, recoverable runs — backed by a **989-test** suite, **62/62** verification checks green.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — evaluation-gated campaign engine

- Attributable campaign engine with immutable evidence models and fail-closed **GO / NO-GO** decisions; recovery restricted to evidence declared before reassessment.
- Verification gates on every agent action: no completion claim without independent provider readback — the eval gate decides, not the agent.

GlacierEQ estate — production agent systems

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Per-task latency instrumentation and decision audit logs across agent runs — the tracing practice that makes benchmarking honest.

SKILLS

Languages: Python · TypeScript · SQL · Bash · JavaScript / Node.js

AI / Field: agentic workflows · AI governance · model evaluation · context management · MCP servers & connectors · eval harnesses · quality scoring · LLM-as-a-judge · golden datasets · trace intelligence · parallel tool orchestration · function calling · structured outputs · guardrails · fail-closed recovery

Production: full lifecycle ownership · root-cause diagnosis · fail-closed deploy gates · JSON Schema / Pydantic-style validation · HTTP APIs & automation · CI regression gates · observability & run logging · latency/cost tradeoff analysis · cross-layer debugging · documentation & handover

Evidence boundary: portfolio repositories are independent work. No GitLab-product internals, Ruby on Rails production, or Git internals depth claimed. No customer logos, no production-at-GitLab claims. GitHub is the forge of record.